Noida Institute of Engineering and Technology, Greater Noida

Expectations and Probability Distribution

Module 4

Subject Name: Statistics and Probability

Subject Code: BAS0303N

Dr. Lokesh chaudhary

Department of Mathematics

B Tech 3rd Sem

Syllabus

Module 1: (Statistical Techniques-I)

Introduction: Measures of central tendency: Mean, Median, Mode, Standard deviation, Quartile deviation, Moment, Skewness, Kurtosis.

Module 2: (Statistical Techniques-II)

Curve Fitting, Method of least squares, fitting of straight lines, Fitting of second-degree parabola, Exponential curves, Correlation and Rank correlation, Linear regression, nonlinear regression and multiple linear regression.

Module 3: (Probability and Random Variable)

Random Variable: Definition of a Random Variable, Discrete Random Variable, Continuous Random Variable, Probability mass function, Probability Density Function, Distribution functions.

Multiple Random Variables: Joint density and distribution Function, Properties of Joint Distribution function, Marginal density Functions, Conditional Distribution and Density, Statistical Independence, Central Limit Theorem (Proof not expected).

Syllabus

Module 4: (Expectations and Probability Distribution)

Expectations of single Random Variable, Mean, Variance, Moment Generating Function, Binomial, Poisson, Normal, Exponential distribution.

Module 5: (Hypothesis Tests and Control Charts)

Testing a Hypothesis, Null hypothesis, Alternative hypothesis, Level of significance, Confidence limits, Test of significance of difference of means, Z-test, t-test and Chi-square test, F-test, One way ANOVA.

Statistical Quality Control (SQC), Control Charts, Control Charts for variables (Mean and Range Charts), Control Charts for Variables (p, np and C charts).

Branch Wise Application

- Data Analysis
- Artificial intelligence
- Network and Traffic modeling

Course Objectives

- The objective of this course is to familiarize the students with concepts of Probability and statistical techniques. It aims to equip the students with adequate Knowledge of statistics that will enable them in formulating Problems and solving problems analytically.

  The students will learn:
- Understand the concept of correlation, moments, skewness and kurtosis and curve fitting.
- Apply the concept of hypothesis testing and statistical quality control to create control charts.
- Remember the concept of probability to evaluate probability distributions.
- Understand the concept of Mathematical Expectations and Probability Distribution.

Course Outcomes

CO1: Apply the concept of moments, skewness and kurtosis in relevant field.

CO2: Apply the concept of correlation, regression and curve fitting with real world problems.

CO3: Apply the concept of probability and random variable.

CO4: Apply the concept of Mathematical Expectations and Probability Distribution in real life problems.

CO5: Apply the concept of hypothesis testing and statistical quality control to create control charts.

Prerequisite

- Knowledge of Maths 1 B.Tech.
- Knowledge of Maths 2 B.Tech.
- Knowledge of Permutation and Combination.

Module Objective(CO4)

1. A basic knowledge in **probability theory**.
2. The student is able to reflect developed mathematical methods in **probability** and **statistics**.
3. Understand the concept of Probability and its usage in various business applications.
4. The **binomial distribution** model allows us to compute the **probability** of observing a specified number of "successes" when the process is repeated a specific number of times
5. **Poisson Distribution** is a tool that helps to predict the probability of certain events from happening when you know how often the event has occurred.
6. To learn the characteristics of a typical **normal curve**.
7. To explore the key properties, such as the moment-generating function, mean and variance, of a **normal** random variable.

Module Content

- Expectations: Introduction
- Expected Value of a Random Variable
- Mean,
- Variance,
- Moment Generating Function,
- Binomial,
- Poisson,
- Normal,
- Exponential distribution.

Topic objective (CO4)

Mathematical expectation

- To get a general understanding of the mathematical expectation of a discrete random variable.
- To learn and be able to apply a shortcut formula for the variance of a discrete random variable.
- To be able to calculate the mean and variance of a linear function of a discrete random variable.

Mathematical expectation(CO4)

Mathematical expectation or expected value of a random variable:

- **When variable is discrete random variable:** The expected value of a discrete random variable is a weighted average of all possible values of the random variable, where the weights are the probabilities associated with the corresponding values. It is denoted by E(x)

If x denotes a discrete random variable which assumes values $x_1, x_2 .. \, x_n$ with corresponding probabilities $p_1, p_2 \ldots.. \, p_n$ respectively, then

$$E(x) = x_1 p_1 + x_2 p_2 + \cdots + x_n p_n = \sum_{i=1}^{n} p_i x_i = \sum px$$

Where $\sum p = 1$

Mathematical expectation(CO4)

rth Moment about origin in terms of expectation is written as $\mu_r = E(x-\bar{x})^r$

$\boldsymbol{Mean} = E(x) = \frac{\sum px}{\sum p} = \sum px$ because $\sum p = 1$

So E(x) represents the mean

Variance $\boldsymbol{\mu_2} = E[\{x-E(x)\}^2] = E(x^2) - [E(x)]^2$

- **When variable is continuous:** If x is a continuous random variable then expectation E(x) is written as

$$E(x) = \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{\infty} xdF(x)$$

Where f(x) is probability density function.

Mathematical expectation(CO4)

Laws of Expectations :

- **Theorem 1. If C is a constant then E(C)=C.**

Proof: Function is constant so assigns value is C to each value within its domain , we have

$P(C = C) = 1$

$P(C = D) = 0, D \neq C$

Therefore $E(C) = C.1 + D.0 = C.$

Hence proved

- **Theorem 2. If a is a constant, then $\boldsymbol{E(aX) = aE(X)}$.**

Proof: Let X takes values $x_1, x_2..$ with corresponding probabilities $p_1, p_2 \ldots..$ respectively, then

$E(aX) = a\, x_1 p_1 + a\, x_2 p_2 + \cdots = a(x_1 p_1 + x_2 p_2 + \cdots) = aE(X)$

Mathematical expectation(CO4)

- **Theorem 3. Expectation of the sum of two random variables**

If X and Y are two discrete random variables with finite expectations E(X) and E(Y) respectively , then the expectation of their sum exists and is the sum of their expectations i.e.

$$E(X + Y) = E(X) + E(Y)$$

Generalization: $E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$

Or $E(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} E(X_i)$

Note : if a and b are constant then $E(aX + b) = aE(X) + b$

- **Theorem 4.**Expectation of the product of two independent random variables i.e. E(XY) = E(X)E(Y)

Generalization:$E(X_1, X_2, \dots X_n) = E(X_1)E(X_2) \dots E(X_n)$

Mathematical expectation(CO4)

❖ **Theorem 5.** Expectation of the difference of two random variables,

i.e. E(X-Y)=E(X)-E(Y)

Q1. What is the expected value of the number of the number of points that will be obtained in a single throw with an ordinary die? Find variance also.

Solution. Here the variate is the number of points showing on a die. It assumes the values 1,2,3,4,5,6 with probability $\frac{1}{6}$ in each case.

$$E(x) = p_1x_1 + p_2x_2 + \ldots + p_6x_6 = \frac{1}{6}\cdot 1 + \frac{1}{6}\cdot 2 + \cdots\cdots + \frac{1}{6}\cdot 6 = 3.5$$

$$\mathrm{var}(x) = \mathrm{E}(x^2) - [\mathrm{E}(x^2)] = \frac{1}{6}(1^2 + 2^2 + \cdots\cdots + 6^2) - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

Daily Quiz(CO4)

Q1. What is the mathematical expectation of the sum of points on n dice?

Q2. Let X denote the number of women in the interview pool. We have found the probability mass function of X.

$X = x$	0	1	2
$P(x)$	$2/11$	$5/11$	$4/11$

How many women do you expect in the interview pool?

Recap(CO4)

- ✓ Mathematical expectation
- ✓ Mean
- ✓ Variance

Topic Objective(CO4)

In probability theory and statistics, the *moment-generating function* of a real-valued random variable is an alternative specification of its probability .

Moment Generating Function(CO4)

❑ MOMENT GENERATING FUNCTION:

An indirect method for computing moments is known as moment generating function. The method depends on the finding of the moments generating function.

❖ **In Case of Continuous Variable x,it is defined as**

$$M(t) = \int_a^b e^{tx} f(x)dx$$

Where integral is a function of parameter t only. The limit a, b can be $-\infty$ and ∞ respectively. It is possible to associate a moment generating function with the distribution only when all the moments of the distribution are finite.

Let us see how $M(t)$ generates moments. For this let us assume

Moment Generating Function(CO4)

That $f(x)$ is a distribution function for which the integral given by(1) exsists. Then e^{tx} may be expanded in a power series and the integration may be performed term by term. It follows that

$$M(t) = \int_a^b \left(1 + tx + \frac{t^2}{2}x^2 + \cdots\right) f(x)dx$$

$$= \int_a^b f(x)dx + t\int_a^b xf(x)dx + \ldots.$$

$$= v_0 + v_1 t + v_2 . \frac{t^2}{2} + \cdots$$

Obviously, the coefficient of $\frac{t^r}{r}$ in (2) is the r^{th} moment about the origin.

Also, $\left|\frac{d^r}{dt^r} M(t)\right|_{t=0} = \left|\frac{v_r}{r} r + v_{r+1}t + \cdots\right|_{t=0} = v_r$, Thus v_r about origin $= r^{th}$ derivative of $M(t)$ with $t = 0$.

Moment Generating Function(CO4)

Although the moment generating function (m.g.f) has been defined for the variable x e.g if $z = x - m(m\ is\ mean)$,the r^{th} moment about z will r^{th} moment about z will given r^{th} moment of x about the mean m.

$$M_z(t) = M_{x-m}(t) = \int_a^b e^{tx} f(x)dx = e^{-m}\ M_x(t)$$

❖ **In case of discrete distribution:**

The moment generating function is given by

$$M_z(t) = \sum e^{tz} P$$

By expanding e^{tz} we have

Moment Generating Function(CO4)

$$M_z(t) = \sum \left(1 + tz = \frac{t^2}{2} z^2 + \cdots \right) P$$

$$= \sum P + t \sum P z + \frac{t^2}{2} \sum P z^2 + \cdots$$

$$= v_0 + v_1 t + v_2 . \frac{t^2}{2} + \cdots$$

We get

$$v_r = \left| \frac{d^r}{dt^r} M(t) \right|_{t=0}$$

Moment generating function about any arbitrary number

$$M_{x-a}(t) = E\left(e^{t(x-a)}\right) = e^{-at} M_x(t)$$

Properties for moment generating function:

1. The moment generating function of the sum of two independent chance variables is the product of their respective moment generating function, i.e $M_{x+y}(t) = M_x(t) \times M_y\,(t)$
2. Effect of change of origin and scale on m.g.f.

$$M_u(t) = e^{-at/h} M_x\left(t/h\right)$$

Q1. find the moment generating function of the discrete binomial distribution given by $P(x) = {}^{n}C_x p^x q^{n-x}$ where $q = 1 - p$

Also find the 1st and 2nd moment about the mean .

Solution: Moment generating function given about the origin is given by

Moment Generating Function(CO4)

$$M_x(t) = \sum e^{xt} P = \sum e^{tx\,n}C_x p^x q^{n-x}$$

$$= \sum {}^nC_x (e^t p)^x q^{n-x}$$

$$= (q + pe^t)^n$$

$$v_1 = \left|\frac{d}{dt} M(t)\right|_{t=0} = [n(q + pe^t)^{n-1} pe^t]_{t=0} = np$$

$$v_2 = \left|\frac{d^2}{dt^2} M(t)\right|_{t=0}$$

$$= [np\{e^t(n-1)(q + pe^t)^{n-2} pe^t + (q + pe^t)^{n-1} e^t]_{t=0}$$

Moment Generating Function(CO4)

$= [np(q + pn)]$
$= npq + n^2p^2$

Hence first and second moments about the mean are given by

$\mu_1 = 0$

Since $\bar{x} = v_1 = np$

$\therefore \ \mu_2 = v_2 - \bar{x}^2$

$\mu_2 = v_2 - {v_1}^2$

$= npq + n^2p^2 - n^2p^2$

$= npq$

Hence, mean= np, S.D. $= \sqrt{\mu_2} = \sqrt{npq}$.

Moment Generating Function(CO4)

Q2. find the moment generating function of the discrete Poisson distribution given by $P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$. Also find the first and second moments about the mean.

Solution: Moment generating function about the origin is given by

$$M_x(t) = \sum e^{xt} P = \sum e^{tx} \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda}\, e^{\lambda e^t} = e^{\lambda(e^t - 1)}$$

$$v_1 = \left|\frac{d}{dt} M(t)\right|_{t=0} = [e^{\lambda(e^t-1)}\lambda e^t]_{t=0} = \lambda$$

$$v_2 = \left|\frac{d^2}{dt^2} M(t)\right|_{t=0} = \lambda(\lambda + 1)$$

Moment Generating Function(CO4)

Hence , the first and second moments about mean are given by

$\mu_1 = 0$

Since $\bar{x} = \nu_1 = \lambda$

$\therefore\ \mu_2 = \nu_2 - \bar{x}^2$

$\mu_2 = \nu_2 - {\nu_1}^2$

$= \lambda(\lambda + 1) - \lambda$

$= \lambda(\lambda + 1)$

Q3. find the m.g.f. of the continuous normal distribution given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} ; -\infty < x < \infty$$

Solution: Moment generating function about the origin is defineds

Moment Generating Function(CO4)

$$M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x)dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t\sigma z} e^{-\frac{1}{2}z^2} dz; \text{ where z} = \frac{x-\mu}{\sigma}$$

$$= \frac{1}{\sqrt{2\pi}} e^{\left(\mu t+\frac{1}{2}t^2\sigma^2\right)} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z-t\sigma)^2} dz$$

$$= e^{\left(\mu t+\frac{1}{2}t^2\sigma^2\right)}.1$$

$$= e^{\left(\mu t+\frac{1}{2}t^2\sigma^2\right)}$$

Daily Quiz(CO4)

Q1. Find the moment generating function about origin of a random variable X whose probability density function is given by

$$f(x) = \frac{1}{2}e^{-|x|}.$$

Q2. Find the moment generating function of x for f(x) = 1, where 0 <x< 1, and thereby confirm that E(x) = 12 and V (x) = 112 .

Weekly Assignments (CO4)

1. Find the moment generating function about origin of random variable X whose p.m.f is given by $p(x) = pq^{x-1}, x = 1,2,3 \ldots \ldots$ Hence find mean and variance of the distribution.
2. Find the mean and variance of Binomial distribution.
3. Find the mean and variance of normal distribution.

Recap(CO4)

- ✓ Mathematical expectation
- ✓ Mean
- ✓ Variance
- ✓ Moment Generating Function

Topic Objective(CO4)

The **probability distributions** are very much helpful for making predictions. Estimates and predictions form an **important** part of research investigation. With the help of Probability distributions, we make estimates and predictions for the further analysis.

Probability distributions(CO4)

Theoretical probability distribution

- Discrete probability distributions
  - Binomial Distribution
  - Poisson Distribution
- Continuous probability distributions
  - Normal Distribution
  - t- Distribution
  - F-Distribution

Binomial Distribution(CO4)

Binomial Probability Distribution: Probability distribution defined as follows is known as binomial Probability distribution.

$$P(X = r) = {}^{n}C_{r} p^{r} q^{n-r}, r = 1,2 \dots n$$

Where n is no of trial which are finite ,r be the success in n trials and $p + q = 1$, p is probability of success and q is probability of failure.

Assumptions For Binomial distribution:

- n, the number of trials is finite
- Each trial has only two possible outcomes usually called success and failure.
- All trials are independent.
- p and q is constant for all trials.

Recurrence or recursion formula:

$P(r) = {}^{n}C_{r} p^{r} q^{n-r} = \frac{n!}{r!(n-r)!} p^{r} q^{n-r}$….(1)

Equation (1) denote binomial distribution.

$P(r+1) = {}^{n}C_{r+1} p^{r+1} q^{n-r-1} = \frac{n!}{(r+1)!(n-r-1)!} p^{r+1} q^{n-r-1}$….(2)

By equation (1) and (2)

$$\frac{P(r+1)}{P(r)} = \frac{(n-r)!}{(n-r-1)!} \times \frac{r!}{(r+1)!} \times \frac{p}{q}$$

$P(r+1) = \frac{(n-r)}{(r+1)} \frac{p}{q} . P(r)$…..(3)

Equation (3) is known as Recurrence Formula.

Mean Of Binomial distribution:

$$\boldsymbol{\mu = \sum_{r=0}^{n} rP(r)}$$

For Binomial distribution $P(r) = {}^{n}C_r p^r q^{n-r}$

$$\mu = \sum_{r=0}^{n} r\,{}^{n}C_r p^r q^{n-r}$$

By expanding we have

$$\Rightarrow 0 + 1.\,{}^{n}C_1 pq^{n-1} + 2.\,{}^{n}C_2 p^2 q^{n-2} + \cdots + n.\,{}^{n}C_n p^n$$
$$\Rightarrow np[{}^{n-1}C_0 q^{n-1} + {}^{n-1}C_1 pq^{n-2} + \cdots + {}^{n-1}C_{n-1} p^{n-1}]$$
$$\Rightarrow np(q+p)^{n-1} = np$$

Hence mean of binomial distribution is np.

Variance of Binomial Distribution:

$$Variance\ \sigma^2 = \sum_{r=0}^{n} r^2 P(r) - \mu^2$$

$$= \sum_{r=0}^{n} [r + r(r-1)]P(r) - \mu^2$$

$$= \sum_{r=0}^{n} rP(r) + \sum_{r=0}^{n} r(r-1)P(r) - \mu^2$$

$$= np + \sum_{r=0}^{n} r(r-1)P(r) - n^2p^2$$

$$= np + n(n-1)p^2 - n^2p^2 = np(1-p) = npq$$

Cont…(CO4)

Hence the **Variance of binomial distribution is $\boldsymbol{npq}$** and Standard deviation is $\sqrt{npq}$.

Moment generating function of binomial Distribution:

i. About origin

$$M_x(t) = E(e^{xt}) = \sum_{x=0}^{n} {}^{n}C_x (pe^t)^x q^{n-x} = (q + pe^t)^n$$

ii. About mean

$$M_{x-np}(t) = E\left(e^{t(x-np)}\right) = e^{-np}\ E(e^{xt}) = e^{-np}\ M_x(t) = e^{-npt}(q + pe^t)^n$$
$$= (qe^{-pt} + pe^{qt})^n$$

Applications of Binomial Distribution:

1. In problem concerning no. of defectives in sample production line.
2. In estimation of reliability of systems.
3. No. of rounds fired from a gun hitting a target.
4. In radar detection.

Q1. If 10% of bolts are produced by a machine are defective , determine the probability that out of 10 bolts chosen at random

i. 1
ii. None
iii. At most 2 bolts will be defective

Problem's based on Binomial Distribution(CO4)

Solution: let p and q are the probability of defective and non defective bolts respectively.

$p = \frac{10}{100} = \frac{1}{10}, q = 1 - \frac{1}{10} = \frac{9}{10}$ and n=10 (no of bolts chosen)

The Probability of r defective bolts out of n bolt chosen at random is given by

$P(r) = {}^{n}C_r p^r q^{n-r}$

i. Here r=1,

$$P(1) = {}^{10}C_1 \left(\frac{1}{10}\right)^1 \left(\frac{9}{10}\right)^{10-1} = 0.3874$$

ii. Here r=0

$$P(0) = {}^{10}C_0 \left(\frac{1}{10}\right)^0 \left(\frac{9}{10}\right)^{10-0} = 0.3486$$

Problem's based on Binomial distribution(CO4)

iii.Prob.that at most 2 bolts will be defective $=P(\leq 2) = P(0) + P(1) + P(2)$

$$P(2) = {}^{10}C_2 \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^{10-2}$$

$=45\left(\frac{1}{100}\right)(0.43046) = 0.1937$

From(4).Required Probability$=P(0) + P(1) + P(2)$

$$= 0.3486 + 0.3874 + 0.1937 = 0.9297$$

Q2. Out of 800 families with 4 children each, how many families would be expected to have

i. 2 boys and 2 girls ii. At least one boy iii no girl

iv. Atmost two girls? Assume equal probability for boys and girls.

Cont…(CO4)

Solution: Probability for boys and girls are equal

$p = probability\ of\ having\ a\ boy = \frac{1}{2},$

$q = probability\ of\ having\ a\ girl = \frac{1}{2}$ n=4 N=800

i. The expected number of families having 2 boys and 2 girls

$$= 800\,{}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 300$$

ii. The expected number of families having at least one boy

$$= 800\left[{}^4C_1 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 + {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 + {}^4C_3 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + {}^4C_4 \left(\frac{1}{2}\right)^4\right]$$

$$= 750$$

Cont…(CO4)

iii. The expected number of families having no girl i.e. having 4 boys = $800\left[{}^{4}C_{4}\left(\frac{1}{2}\right)^{4}\right] = 50$

iv. The expected number of families having almost two girls i.e. having at least 2 boys

$$= 800\left[{}^{4}C_{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{2} + {}^{4}C_{3}\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{3} + {}^{4}C_{4}\left(\frac{1}{2}\right)^{4}\right] = 550$$

Daily Quiz(CO4)

1. Four persons in a group of 20 are graduates. If 4 persons are selected at random from 20, find the probability that all 4 are graduates.

   Ans: 0.0016

2. The Prob. that a bulb produced by a factory will fuse after use of 150 days is 0.05. Find the probability that out of 5 such bulbs at least one bulb will fuse after use of 150 days of use.

   Ans: $1 - \left(\frac{19}{20}\right)^5$

Recap (CO4)

- ✓ Mathematical expectation
- ✓ Mean
- ✓ Variance
- ✓ Moment Generating Function
- ✓ Binomial distribution

Poisson Distribution(CO4)

Poisson distribution: Probability distribution defined as follows is known as Poisson Probability distribution.

$$P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}, (r = 0,1,2,3 \dots .)$$

Where $\lambda\ finite\ number = np$.

Recurrence formula for Poisson Distribution:

Poisson distribution $P(r) = \frac{e^{-\lambda}\lambda^r}{r!} \dots \dots \dots (1)$

$$P(r+1) = \frac{e^{-\lambda}\lambda^{r+1}}{(r+1)!} \dots\dots\dots (2)$$

$$\frac{P(r+1)}{P(r)} = \frac{\lambda r!}{(r+1)!} = \frac{\lambda}{r+1}$$

$$P(r+1)=\frac{\lambda}{r+1}P(r), r=01,2,3\ldots\ldots$$

This is called the recurrence or recursion formula for Poisson distribution.

Mean of the Poisson distribution:

Mean $\mu=\sum_{r=0}^{\infty} rP(r)$

$$=\sum_{r=0}^{\infty} r.\frac{e^{-\lambda}\lambda^r}{r!}$$

$$=e^{-\lambda}\sum_{r=1}^{\infty}\frac{\lambda^r}{(r-1)!}$$

Cont…(CO4)

$$= e^{-\lambda}\left(\lambda + \frac{\lambda^2}{1!} + \frac{\lambda^3}{2!} + \cdots.\right)$$

$$= \lambda e^{-\lambda}\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots.\right)$$

$$= \lambda e^{-\lambda}\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots.\right)$$

$$= \lambda e^{-\lambda} e^{\lambda} = \lambda$$

Mean $= \lambda$ for Poisson distribution.

Variance of Poisson Distribution:

$$Variance\ \sigma^2 = \sum_{r=0}^{\infty} r^2 P(r) - \mu^2$$

$$= \sum_{r=0}^{\infty} r^2 \frac{e^{-\lambda}\lambda^r}{r!} - \mu^2$$

$$= e^{-\lambda} \sum_{r=0}^{\infty} \frac{r^2 \lambda^r}{r!} - \lambda^2$$

$$= e^{-\lambda} \left[\frac{1^2 \lambda^1}{1!} + \frac{2^2 \lambda^2}{2!} + \frac{3^2 \lambda^3}{3!} + \cdots \right] - \lambda^2$$

$$= \lambda e^{-\lambda} \left[1 + \frac{2\lambda}{1!} + \frac{3\lambda^2}{2!} + \cdots \right] - \lambda^2$$

$$= \lambda e^{-\lambda} \left[1 + \frac{(1+1)\lambda}{1!} + \frac{(1+2)\lambda^2}{2!} + \cdots \right] - \lambda^2$$

$$= \lambda e^{-\lambda} \left[\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots \right) + \left(\frac{\lambda}{1!} + \frac{2\lambda^2}{2!} + \cdots \right) \right] - \lambda^2$$

$$= \lambda e^{-\lambda}\left[e^{\lambda} + \lambda\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right)\right] - \lambda^2$$
$$= \lambda e^{-\lambda}[e^{\lambda} + \lambda e^{\lambda}] - \lambda^2$$
$$= e^{\lambda}e^{-\lambda}\lambda[1 + \lambda] - \lambda^2$$
$$= \lambda[1 + \lambda] - \lambda^2$$
$$= \lambda$$

Hence , the Variance of the Poisson distribution is also λ.

Applications of Poisson Distribution:

i. Arrival pattern of the defective vehicles in a workshop.

ii. Patients in hospitals.

iii. Telephone calls.

iv. Emission of radioactive (α) particles.

Problem based on Poisson Distributions(CO4)

Q1. If the Variance of the Poisson distribution is 2 , find the probability for r=1,2,3,4 from the recurrence relation of the Poisson Distribution. Also find $P(r \geq 4)$.

Solution: Given that Variance $= 2 = \lambda$

Recurrence relation for Poisson distribution

$$P(r+1) = \frac{\lambda}{r+1} P(r) = \frac{2}{r+1} P(r) \ldots \ldots (1)$$

Poisson Distribution $P(r) = \frac{e^{-\lambda}\lambda^r}{r!}$

So $P(r) = \frac{e^{-2}2^r}{r!} \Longrightarrow P(0) = \frac{e^{-2}2^0}{0!} \Longrightarrow e^{-2} = 0.1353$

Now putting r=0,1,2,3 in equation (1)

Cont…(CO4)

$$P(1) = 2P(0) = 2 \times 0.1353 = 0.2706$$

$$P(2) = \frac{2}{2}P(1) = \frac{2}{2} \times 0.2706 = 0.2706$$

$$P(3) = \frac{2}{3}P(2) = \frac{2}{3} \times 0.2706 = 0.1804$$

$$P(4) = \frac{2}{4}P(3) = \frac{1}{2} \times 0.1804 = 0.0902$$

Now to calculate $P(r \geq 4), we\ have$

$$P(r \geq 4) = 1 - [P(0) + P(1) + P(2) + P(3)]$$

$$= 1 - [0.1353 + 0.2706 + 0.2706 + 0.1804]$$

$$= 0.1431$$

Q2. Fit a Poisson distribution to the following data and calculate theoretical frequencies.

Cont…(CO4)

Deaths:	0	1	2	3	4
Frequencies	122	60	15	2	1

Mean of Poisson distribution $\lambda = \frac{\sum fx}{\sum f}$

$$= \frac{0 \times 122 + 1 \times 60 + 2 \times 15 + 3 \times 2 + 4 \times 1}{122 + 60 + 15 + 2 + 1} \sum f = \mathrm{N} = 200$$

$$= 0.5$$

Required Poisson distribution

$$= N.\frac{e^{-\lambda}\lambda^r}{r!} = 200\,\frac{e^{-0.5}(0.5)^r}{r!} = (121.306)\,\frac{(0.5)^r}{r!}$$

Cont…(CO4)

r	N.P(r)	Theoretical frequencies
0	$(121.306)\dfrac{(0.5)^0}{0!} = 121.306$	121
1	$(121.306)\dfrac{(0.5)^1}{1!} = 60.653$	61
2	$(121.306)\dfrac{(0.5)^2}{2!} = 15.163$	15
3	$(121.306)\dfrac{(0.5)^3}{3!} = 2.527$	3
4	$(121.306)\dfrac{(0.5)^4}{4!} = 0.3159$	0
		Total=200

Recap (CO4)

- ✓ Mathematical expectation
- ✓ Mean
- ✓ Variance
- ✓ Moment Generating Function
- ✓ Binomial distribution
- ✓ Poisson distribution

Topic objective of Normal Distribution(CO4)

- To define the probability density function of a **normal** random variable.
- To learn the characteristics of a typical **normal curve**.
- To explore the key properties, such as the moment-generating function, mean and variance, of a **normal** random variable.

Normal Distribution(CO4)

Normal Distribution: The general equation of the normal distribution is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Basic properties for Normal distributions:

i. $f(x) \geq 0$

ii. $\int_{-\infty}^{\infty} f(x)dx = 1$

i.e. the total area under the normal curve above x-axis is 1.

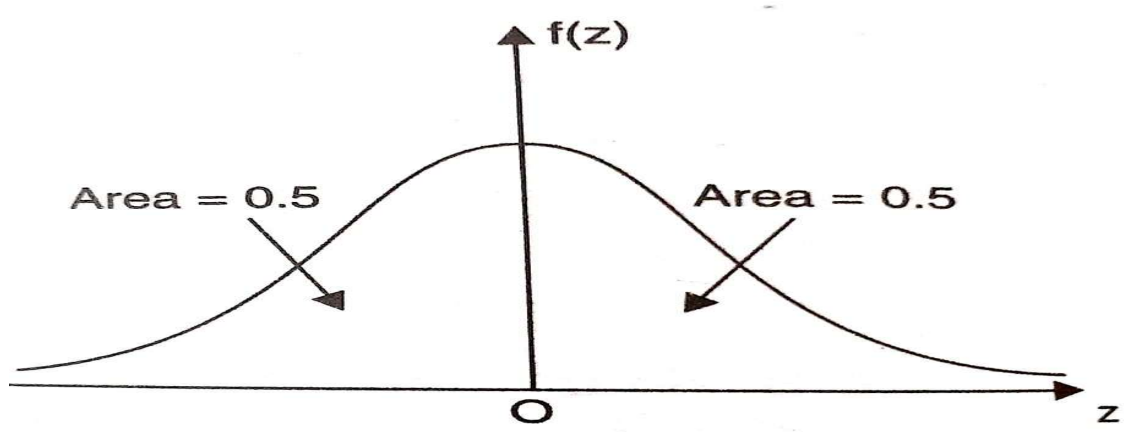
iii. Normal curve is symmetrical about its mean.

iv. The mean, mode and median coincide for this distribution.

Standard form of the normal distribution: The probability density function for the normal distribution in standard form is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

By taking $z = \frac{x-\mu}{\sigma}$, standard normal curve is formed . The total area under the curve is 1 and it is divided into two parts by z=0.



Mean and Variance of Normal distribution(CO4)

Mean of Normal Distribution: A.M of continuous distribution f(x) is given by

$$A.M(\bar{x}) = \frac{\int_{-\infty}^{\infty} xf(x)dx}{\int_{-\infty}^{\infty} f(x)dx}$$

$\bar{x} = \int_{-\infty}^{\infty} xf(x)dx$ because $\int_{-\infty}^{\infty} f(x)dx = 1$

So $\bar{x} = \int_{-\infty}^{\infty} x\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$

Let $\frac{x-\mu}{\sigma} = z$ so that $x = \mu + \sigma z$ therefore dx $= \sigma dz$

$$\bar{x} = \int_{-\infty}^{\infty} (\mu + \sigma z)\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}z^2}\sigma dz$$

$$= \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{1}{2}z^2} dz$$

Because $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz=1$ so in above equation

$$= \mu + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{1}{2}z^2} dz$$

$$= \mu + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} d\left(\frac{z^2}{2}\right)$$

$$= \mu + \frac{\sigma}{\sqrt{2\pi}} \left(e^{-\frac{1}{2}z^2}\right)_{-\infty}^{\infty}$$

$\bar{x} = \mu$

Variance of Normal distribution :

$$= \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx$$

$$= \int_{-\infty}^{\infty} x^2 f(x) dx \ -\bar{x}^2 \ \dots\dots\dots\dots (1)$$

Let I $= \int_{-\infty}^{\infty} x^2 f(x) dx \ = \int_{-\infty}^{\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\bar{x}}{\sigma}\right)^2} dx$

$\frac{x-\bar{x}}{\sigma} =$ z so that $x = \bar{x} + \sigma z$ therefore d$x = \sigma dz$

$$I = \int_{-\infty}^{\infty} (\bar{x} + \sigma z)^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \sigma dx$$

$$= -\frac{\sigma^2}{\sqrt{2\pi}} \left(z e^{-\frac{1}{2}z^2}\right)_{-\infty}^{\infty} + \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2}\, dx + \bar{x}^2$$

$$= 0 + \sigma^2 + \bar{x}^2$$

$$= \sigma^2 + \bar{x}^2$$

In equation(1)

Variance $= \sigma^2 + \bar{x}^2 - \bar{x}^2 = \sigma^2$

s.d. of normal distribution is σ.

Problem based on Normal distribution(CO4)

Q1. A sample of 100 dry battery cells tested to find the length of the life produced the following results :

$$\bar{x} = 12\ hours, \sigma = 3\ hours$$

Assuming the data to be normally distributed, what percentage of battery cells are expected to have life

i. More than 15 hours

ii. Less than 6 hours

iii. between 10 and 14 hours

Solution: x denotes the life of dry battery cells .

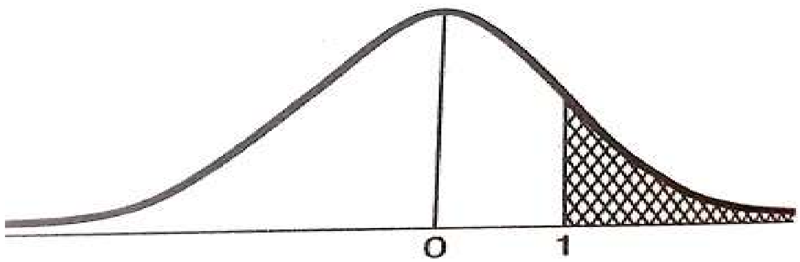
And $z = \frac{x-\bar{x}}{\sigma} = \frac{x-12}{3}$

i. When x = 15 then z = 1

Therefore $P(x > 15) = P(z > 1)$

$= P(0 < z < \infty) - P(0 < z < 1)$

$= 0.5 - 0.3413 = 0.1587 = 15.87\%$



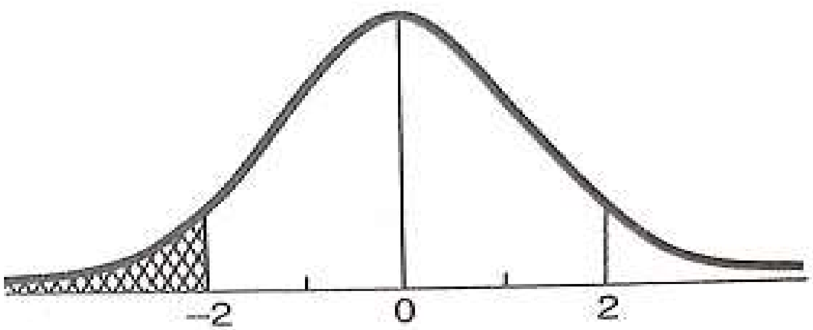
ii. When $x = 6$ then $z = -2$

Therefore

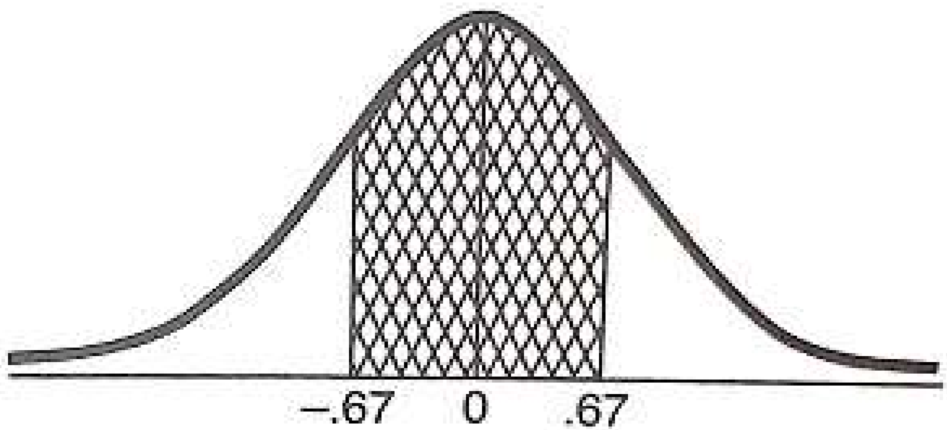
$P(x < 6) = P(z < -2) = P(z > 2)$

$= P(0 < z < \infty) - P(0 < z < 2)$

$= 0.5 - 0.4772 = 0.0228 = 2.28\%$



iii. When $x = 10$ then $z = -0.67$
when $x = 14$ then $z = 0.67$
$P(10 < x < 14) = P(-0.67 < z <).67)$
$= 2P(0 < z < 0.67) = 2 \times 0.2485 = 49.70\%$



Daily Quiz(CO4)

1. In a distribution exactly Normal, 31% of the items are under 45 and 8% are over 64. What are the mean and Standard deviation of this Distribution? It is given that if

$f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{\frac{-x^2}{2}} dx$ then $f(0.5) = 0.19, f(1.4) = 0.42.$

2. The lifetimes, T hours, of a certain brand of battery are assumed to be Normally distributed with a mean of 825 and a variance of 900 . Find the probability that one such battery will last between 801 and 870 hours.

Ans: 0.7213

Recap(CO4)

- ✓ Mathematical expectation
- ✓ Mean
- ✓ Variance
- ✓ Moment Generating Function
- ✓ Binomial distribution
- ✓ Poisson distribution
- ✓ Normal Distributions

Exponential distribution:

A continuous random variable $X = x$ which has the following Pdf

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & for\ x \geq 0 \\ 0 & for\ x < 0 \end{cases}$$

Where λ is a parameter.

is called exponential distribution.

Mean of Exponential distribution: We know that mean is given by

$$\text{Mean} = E(X = x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_{0}^{\infty} x\lambda e^{-\lambda x}\, dx$$

$$= \lambda \int_{0}^{\infty} x e^{-\lambda x}\, dx$$

Exponential Distribution(CO4)

$= \lambda \int_0^{\infty} x^{2-1} \ e^{-\lambda} \ \ dx,$ which is a Gamma function

$$= \lambda \frac{\Gamma 2}{\lambda^2} = \frac{1}{\lambda}$$

$$\left[Mean = \frac{1}{\lambda}\right]$$

Variance of Exponential distribution:

We know that the variance is given by

$$\text{Variance}(X = x) = E(x^2) - (E(x))^2 \ \ldots\ldots\ldots.(1)$$

Now $\quad E(x^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$

$$= \int_0^{\infty} x^2 \lambda e^{-\lambda x} \, dx$$

$= \lambda \int_0^{\infty} x^{3-1} e^{-\lambda x} \, dx$, which is a Gamma function

$$= \lambda \frac{\Gamma 3}{\lambda^3} = \frac{2}{\lambda^2}$$

Exponential Distribution(CO4)

From (1), Variance$(x) = E(x^2) - [E(x)]^2$

$$V(x) = \frac{2}{\lambda^2} - \left[\frac{1}{\lambda}\right]^2 = \frac{1}{\lambda^2}$$

Hence $\mathbf{Variance}(\boldsymbol{x}) = \frac{\mathbf{1}}{\boldsymbol{\lambda}^{\mathbf{2}}}$

Q1. The length of Telephone conversation is an exponential variate with mean 3minutes. Find the Probability that call

(a) End in less than 3 minutes.

(b) End between 3 to 5 minutes.

Solution. Given, Mean$= 3 = \frac{1}{\lambda}$

$$\Rightarrow \lambda = \frac{1}{3}$$

$$P(x < 3) = \int_0^3 f(x)dx = \int_0^3 \lambda e^{-\lambda x}\, dx = \int_0^3 \frac{1}{3} e^{-x/3} dx$$

Exponential Distribution(CO4)

$$= \frac{1}{3}\left[\frac{e^{-x/3}}{-1/3}\right]_0^3 = -(e^{-1} - e^{-0}) = 1 - \frac{1}{e}$$

$$P(3 < x < 5) = \int_3^5 f(x)dx = \int_3^5 \lambda e^{-\lambda x} dx$$

$$= \frac{1}{3}\int_3^5 e^{-x/3} dx = \frac{1}{3}\left[\frac{-x/3}{-1/3}\right]_3^5 = -\left[e^{-5/3} - e^{-1}\right]$$

$$= \frac{1}{e} - \frac{1}{e^{5/3}}$$

Daily Quiz(CO4)

Q1. The length of Telephone conversation is an exponential variate with mean 7 minutes. Find the Probability that call

(a) End in less than 7 minutes.

(b) Ends between 7 to 10 minutes.

Q2. A monitor issues a warning signal when an action is needed as part of a production process. The interval, X hours, between successive signals follows an exponential distribution with parameter 0.08. Find the probability that the interval between the next two signals is Longer than 50 hours.

Ans: 0.0183

Weekly Assignment(CO4)

Q1. The length of Telephone conversation is an exponential variate with mean 5 minutes. Find the Probability that call

(a) End in less than 5 minutes.

(b) tabels between 5 to 10 minutes.

Q2. The experience shows that 4 accidents occur in a plant on an average per month. Calculate the probabilities of less than 3 accidents in a certain month. Use Poisson distribution. (Given $e^{-4}=0.01832$).

Q3. Net profit of 400 companies is normally distributed with a mean profit of Rs. 150 lakhs and a standard deviation of Rs. 20 lakhs. Find the number of companies whose profits(Rs. Lakhs) are between 100 and 138.Also find the minimum profit of top 15% companies. (Area for Z=2.5 ,0.35 and 0.6 are 0.4938,1.04 and 0.2251)

Weekly Assignment(CO4)

Q4. In a sample of 1000 cases, the mean of a certain test is 14 and standard deviation is 2.5. Assuming distribution to be normal find

i. How many students score between 12 and 15?

ii. How many score above 18?

iii. How many score below 8?

iv. How many score 16?

Given that $\phi(z) = \frac{1}{\sqrt{2\pi}}\int_0^z e^{-z^2/2}dz$, $\phi(0.4) = 0.155, \phi(0.8) = 0.2881, \phi(1.6) = 0.445, \phi(0.6) = 0.0225, \phi(1) = 0.341$).

Q5. A large number of measurement is normally distributed with a mean 65.5 inches and S.D. of 6.2 inches. Find the percentage of measurement that fall between 54.8 inches and 68.8 inches. Given that area (at z = 1.73) = 0.4582 and area (at z = .53) = 0.2019 .

Weekly Assignment(CO4)

Q6. In a distribution exactly Normal, 31% of the items are under 45 and 8% are over 64. What are the mean and Standard deviation of this Distribution? It is given that if $f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{\frac{-x^2}{2}} dx$then $f(0.5) = 0.19, f(1.4) = 0.42$

Q7. The life of army shoes is normally distributed with mean 8 months and standard deviation 2 months. If 5000 airs are insured, how many pairs would you expected to need replacement after 12 months?

Q8. Find the mean and variance of the Binomial, Poisson and Normal distribution.

Weekly Assignment(CO4)

Q9. The probability that a pen manufactured by a company will be defective is $1/10$. If 12 such pens are manufactured, find the probability that

i. Exactly two will be defective
ii. At least two will be defective
iii. None will be defective.

Q10. It is given that 2% of the electric bulbs manufactured by a company are defective. Using Poisson distribution find the probability that a sample of 200 bulbs will contain

i. No defective bulb
ii. Two defective bulbs
iii. Atmost three defective bulbs.

Weekly Assignment(CO4)

Q11. In a test on 2000 electric bulbs,it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D. of 60 hours, estimate the number of bulbs likely to burn for

i. More than 2150 hours

ii. Less than 1950 hours

iii. More than 1920 hours but less than 2160 hours

Q12. In a distribution exactly Normal, 7% of the items are under 35 and 89% are under 63. What are the mean and Standard deviation of this Distribution?

Faculty Video Links, Youtube & NPTEL Video Links and Online Courses Details

- **Self made Video Link:**
- https://youtu.be/6pZXCcoeYiU
- https://youtu.be/izT2QpldbnU
- https://youtu.be/UaLNsZQK8fo
- **Suggested Video Links:**
- https://youtu.be/L3wQw0wva3g
- https://youtu.be/n9qpktdFfLU
- https://youtu.be/_Qlxt0HmuOo
- https://youtu.be/YSwmpAmLV2s
- https://youtu.be/KLnGOL_AUgA
- https://youtu.be/cQp_bJdxjWw
- https://youtu.be/geB0A7CPGaQ
- https://youtu.be/zmyh7nCjmsg
- https://youtu.be/ohquDY3fZqk
- https://youtu.be/izGZLnB-mEo
- https://youtu.be/q48uKU_KWas

MCQ s(CO4)

Q1. Suppose that a random variable x has normal distribution with mean 9 and variance 9. Then the value of c such that $P(x > c) = 0.16$ is-(Given that $\phi(1) = 0.34$)

i. 12

ii. 1

iii. 10

iv. None of these

Q2 .The life of army shoes is normally distributed with mean 8 months and standard deviation 2 months. If 5000 pairs are issued, how many pairs would be expected to need replacement after 12 months are-(Given that $P(z \geq 2) = 0.228$)

i. 114

ii. 4886

iii. 115

iv. 4890

MCQ s(CO4)

Q3. Suppose that a book of 600 pages contains 40 printing mistakes. Assume that these errors are randomly distributed throughout the book and x, the number of errors per page has a Poisson distribution then what is probability that 10 pages selected at random will be free of errors

i. 0.54

ii. 0.15

iii. 0.51

iv. None of these

MCQ s(CO4)

Q4. The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped then the probability that exactly two will strike the target is

i. 0.345

ii. 0.145

iii. 0.245

iv. None of these

Q5. For the standard normal variate z mean and variance are-

i. 0,1

$ii.$ μ, σ^2

iii. 1,0

$iv.$ σ^2, μ

Glossary(CO4)

Q1. The distribution of the number of road accidents per day in a city is poisson with mean 4. find the number of days out of 100 days when there will be

i. No accident
ii. At least two accident
iii. At most three accident
iv. Between two and five accident

Pick the correct option from glossary

a. 91
b. 43
c. 39
d. 2

Glossary(CO4)

Q2.In 800 families with 4 children each ,how many families would be expected to have

i. 2 boys and 2 girls
ii. No girl
iii. At least one boy
iv. At most two girls (Assume equal probabilities for boys and girls)

Pick the correct option from glossary

a. 750
b. 550
c. 50
d. 300

University Expected question(CO4)

Q1. Find the mean and variance of the binomial distribution.

Q2. If the probability of hitting a target is 10% and 10 shots are fired independently. What is the probability that the target will be hit at least once?

Q3. If there are 3 misprints in a book of 1000 pages find the probability that a given page will contain

i. No misprint

ii. More than 2 misprints

Q4. If the heights of 300 students are normally distributed with mean 64.5 inches and standard deviation 3.3 inches. Find the height below which 99% of the student lie.

(Given that $\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz\,, \phi(2.327) = 0.49$).

University Expected question(CO4)

Q7. The experience shows that 4 accidents occur in a plant on an average per month. Calculate the probabilities of less than 3 accidents in a certain month. Use Poisson distribution. (Given $e^{-4}=0.01832$).

Q8. As a result of tests on 20,000 electric bulbs manufactured by a company it was found that the life time of a bulb was normally distributed with an average life of 2040 hours and standard deviation of 60 hours. On the basis of the information, estimate the number of bulbs that is expected to burn for

(i) more than 2150 hours, and (ii) less than 1960 hours.

Q9. Write short note on

i. Binomial distribution

ii. Poisson distribution

Q10. Show that mean of binomial distribution is np.

End Semester Question Paper

Printed Page:- 06 | **Subject Code:- BAS0303**

Roll. No:

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY, GREATER NOIDA
(An Autonomous Institute Affiliated to AKTU, Lucknow)
B.Tech
SEM: III - THEORY EXAMINATION (2024- 2025)
Subject: Statistics & Probability

Time: 3 Hours | **Max. Marks: 100**

General Instructions:

IMP: *Verify that you have received the question paper with the correct course, code, branch etc.*

1. *This Question paper comprises of* ***three Sections -A, B, & C.*** *It consists of Multiple Choice Questions (MCQ's) & Subjective type questions.*

2. *Maximum marks for each question are indicated on right -hand side of each question.*

3. *Illustrate your answers with neat sketches wherever necessary.*

4. *Assume suitable data if necessary.*

5. *Preferably, write the answers in sequential order.*

6. *No sheet should be left blank. Any written material after a blank sheet will not be evaluated/checked.*

<u>**SECTION-A**</u> 20

1. Attempt all parts:-

1-a. Wheat crops badly damaged on account of rains is: (CO1, K2) 1

(a) Cyclical movement
(b) Random movement
(c) Secular trend
(d) Seasonal movement

End Semester Question Paper

1-b. Let the average of three numbers be 16. If two of the numbers are 8 and 12, then the remaining number is..... (CO1, K3) 1

(a) 28
(b) 18
(c) 12
(d) 30

1-c. One card is drawn from a standard pack of 52 plying cards. Find the probability that it is either a king or a queen. (CO2, K3) 1

(a) $\frac{1}{13}$
(b) $\frac{2}{13}$
(c) $\frac{3}{13}$
(d) None of these

1-d. If two events A and B are mutually exclusive, then the probability $P(A \cap B)$ is: (CO2, K2) 1

(a) 0
(b) 1
(c) P(A).P(B)
(d) None of these

End Semester Question Paper

1-e.	In binomial distribution probability of success in each trial remains ________.(CO3. K1)	1
	(a) 0	
	(b) 1	
	(c) Constant	
	(d) Not defined	
1-f.	In Normal Distribution, Mean deviation about mean is (CO3, K1)	1
	(a) σ	
	(b) $2\sigma/5$	
	(c) $4\sigma/5$	
	(d) $6\sigma/7$	
1-g.	The area of critical region depends on the size of.... (CO4, K1)	1
	(a) Type I error	
	(b) Type II error	
	(c) Test statistics	
	(d) Sample	

1-h.	In conducting one way analysis of variance --- test statistics would be used. (CO4,K1)	1
	(a) Z	
	(b) T	
	(c) Chi-Square	
	(d) F	

End Semester Question Paper

1-i. If $A=\{1,4,6\}$, $B=\{3,6\}$ and $C=\{3,4,6\}$ then $A\cap(B\cap C)$ is…. (CO5,K3) 1

(a) $\{3,4,5,6\}$

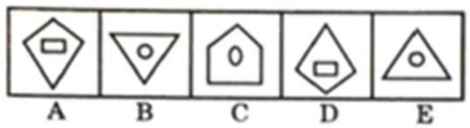
(b) $\{4,6\}$

(c) $\{1,4,6\}$

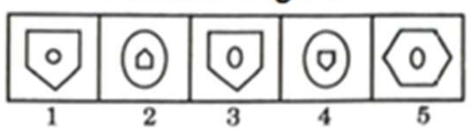
(d) $\{6\}$

1-j. Select a figure from amongst the answer figure which will continue the same series as stablished by the five-figure problem. (CO5,K2) 1

Problem Figure



Answer Figure



(a) 1

(b) 2

(c) 3

(d) 5

End Semester Question Paper

2. Attempt all parts:-

2.a.	Prove that if two variables are independent, their correlation is zero but vice versa is not true. (CO1,K2)	2
2.b.	Find the probability of getting 53 Sundays in a leap year. (CO2,K3)	2
2.c.	A Binomial random variable X satisfies the relation $9P(X=4) = P(X=2)$, When n=6 . Find value of $P(X=1)$. (CO3, K3)	2
2.d.	Define an estimator in statistics. (CO4, K1)	2
2.e.	Check whether the function $f : N \to N$ is defined by $f(x)=x^2+12$ is one-one or not.(CO5, K3)	2

<u>SECTION-B</u> 30

3. Answer any <u>five</u> of the following:-

3-a.	The mean and standard deviation of the marks of 100 candidates was found to be 40 and 5.1, respectively. Later, it was discovered that a score of 40 was wrongly read as 50. Find out the correct mean and standard deviation respectively. (CO1,K3)	6
3-b.	Calculate Spearman's rank correlation coefficient from the following data: (CO1, K3)	6

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

3-c. A random variable X has the following distribution, Find the value of K. Also find Mean and variance (CO2, K3) 6

x	-2	-1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

3-d. A bag X contains 2 white and 3 red balls and another bag Y contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag Y. (CO2, K3) 6

3.e. If 10% of the bolts produced by a machine are defective, determine the probability that out of 10 bolts choosen at random 6

i) 1

ii) None

iii) at most 2 bolts will be defective. (CO3,K3)

3.f. 6

A sample of 20 items has mean 42 units and S.D. 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units. (If the tabular value at 5% LOS for 19 d. f. is 2.09). (CO4, K3)

3.g. How many different words can be formed using all the letters of the world ALLAHABAD 6

1. *When the vowels occupy the even position.*
2. Both L do not occur together. (CO5,K3)

End Semester Question Paper

SECTION-C 50

4. Answer any <u>one</u> of the following:-

4-a. In a partially destroyed laboratory record of analysis of a correlation data, the following results only are legible: (CO1,K3) 10
Variance of x = 9; Regression equations: $8x - 10y + 66=0$, $40x - 18y = 214$.
What were (a) the mean values of x and y (b) the standard deviation of y (c) the coefficient correlation between x and y.

4-b. Fit a second degree parabola by the method of least squares to the following data . (CO1, K3) 10

X	0	1	2	3	4
Y	1	4	10	17	30

5. Answer any <u>one</u> of the following:-

5-a. State and prove Bayes theorem. 10
In a Neighbourhood, 90% children were falling sick due flu and 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, find the child's probability of having flu. (CO2, K3)

End Semester Question Paper

5-b. Let the two dimensional continuous random variable(X,Y) has joint PDF given by 10

$$f(x,y) = \begin{cases} 6x^2y\ , & 0<x<1,\ 0<y<1 \\ 0 & ,\ \text{elsewhere} \end{cases}$$

Find (i) $P(0<x<3/4\ ,\ 1/3<y<2)$ (ii) $P(x + y <1)$. (CO2, K3)

6. Answer any <u>one</u> of the following:-

6-a. A manufacturer of envelopes knowns that the weight of the envelopes is normally distributed with mean 1.9gm and variance 0.01 square gm. Find how many envelopes weighing (i)2gm or more (ii)2.1gm or more, can be expected in a given packet of 1000envelopes? Given that the area under the standard curve between $z=0$ and $z=1$ is 0.3413, between $z=0$ and $z=2$ is 0.4772. (CO3,K3) 10

6-b. 10

Four coins were tossed 200 times. The number of tosses showing 0,1,2,3 and 4 heads were found to be as under.

Fit a binomial distribution to these observed results. Find the expected frequencies. (CO3, K3)

No. of Heads	0	1	2	3	4
No. of Tosses	15	35	90	40	20

End Semester Question Paper

7. Answer any <u>one</u> of the following:-

7-a. 10

To test of significance of the variations of the retail prices in the commodity in three principal cities: Mumbai, Bangalore and Chennai. The four shops were chosen at random in each city and prices observed in INR were as follows:

Mumbai	16	8	12	14
Bangalore	14	10	10	6
Chennai	4	10	8	8

Do the data indicate that the prices in the three cities are significantly different? Given that the tabular value of F is 4.26 5% LOS with d.f. is (2,9). (CO4,K3)

7-b. From the following data, find whether hair color and gender are associated. 10

End Semester Question Paper

Gender↓	*Colour*					
	Fair	**Red**	**Medium**	**Dark**	**Black**	**Total**
Boys	529	849	504	119	36	2100
Girls	544	677	451	97	14	1783
Total	1136	1526	955	216	50	3883

Given that the tabular value of $\chi 2$ is 9.488 at 5%LOS with d.f. 4. (CO4, K3)

8. Answer any one of the following:-

8-a. Solve the following: (CO5,K3) 10

1. What is the sum of all five-digit numbers formed by 2, 3, 4, 5, 6 without repetition?
2. What is the sum of all five-digit numbers formed by 2, 3, 4, 5, 6 with repetition?

End Semester Question Paper

8-b. Study the following table and answer the questions based on it 10

Expenditures of a Company (in Lakh Rupees) per Annum Over the given Years.

Year	Item of Expenditure				
	Salary	Fuel and Transport	Bonus	Interest on Loans	Taxes
1998	288	98	3	23.4	83
1999	342	112	2.52	32.5	108
2000	324	101	3.84	41.6	74
2001	336	133	3.68	36.4	88
2002	420	142	3.96	49.4	98

i) Find the average amount of interest per year which the company had to pay during this period?

ii) The total amount of bonus paid by the company during the given period is approximately what percent of the total amount of salary paid during this period?
iii) Total expenditure on all these items in 1998 was approximately what percent of the total expenditure in 2002?
iv) The total expenditure of the company over these items during the year 2000 is?
v) The ratio between the total expenditure on Taxes for all the years and the total expenditure on Fuel and Transport for all the years respectively is approximately?
(CO5,K3)

Module Recap(CO4)

Students have taught the importance of the following topics….

With the concept of probability to evaluate probability distributions:

- ✓ Mathematical expectation
- ✓ Mean
- ✓ Variance
- ✓ Moment Generating Function
- ✓ Binomial distribution
- ✓ Poisson distribution
- ✓ Exponential Distribution

References (CO4)

Text Books

- N. P. Bali: A Textbook of Engineering Mathematics-IV, University Science Press.
- H. K. Dass: Introduction to Engineering Mathematics, S. Chand.
- S. Ross: A First Course in Probability, 6th Ed., Pearson Education India, 2002.
- W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968.

References (CO4)

Reference Books

- P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003(Reprint).
- R.K. Jain and S.R.K. Iyenger: Advance Engineering Mathematics; Narosa Publishing House, New Delhi.
- J.N. Kapur: Mathematical Statistics; S. Chand & Sons Company Limited, New Delhi.
- D.N.Elhance,V. Elhance & B.M. Aggarwal: Fundamentals of Statistics; Kitab Mahal Distributers, New Delhi.

Thank You